A Comparative Machine Learning Framework for Stellar Subdwarf Classification Using Spectral Metrics and Feature Engineering

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INTRODUCTION

- Cool dwarfs spanning spectral types M0–T9 include low-mass stars and brown dwarfs whose near-infrared spectra are shaped by effective temperature, surface gravity, metallicity, and atmospheric clouds.
- Automated spectral classification is increasingly important for large astronomical surveys, but conventional accuracy may be dominated by the highly represented field-dwarf population.
- Gravity- and metallicity- objects, including *β*, *γ*, d/sd, sd, and esd dwarfs, remain more difficult to distinguish because their spectral characteristics partially overlap.
- This work extends the framework of Zhou et al. [1] by combining binned spectral fluxes with spectral slopes, J- and H-band photometry, and residual features measured relative to field standards of the same spectral subtype.

Figure 1 shows the spectral diversity of field, low-gravity, and metal-poor cool dwarfs. Variations in continuum shape and molecular absorption motivate the engineered spectral features used in this study.

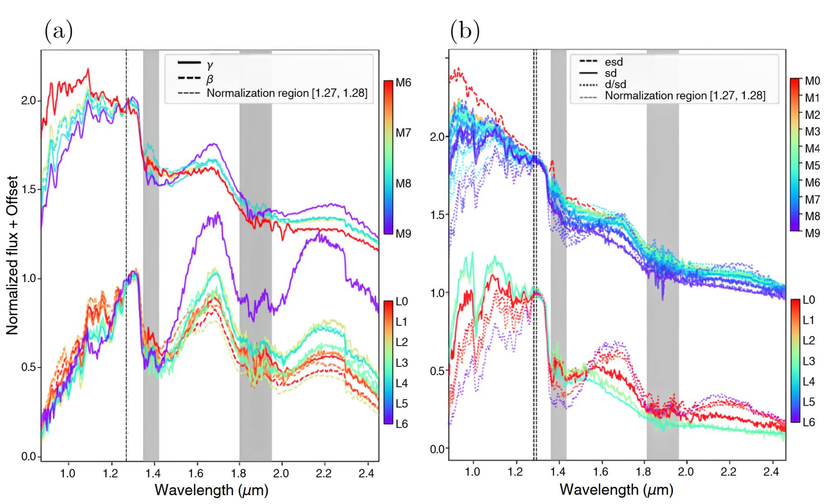


Fig. 1: Spectral diversity of field, low-gravity, and metal-poor M- and L-type standards. Gray regions indicate masked telluric absorption. Adapted from Zhou et al. [1]

Classification targets:

Spectral type: 30 subtypes, M0–T9

Dwarf type: 6 classes — field, *β*, *γ*, d/sd, sd, esd

METHODS

The study used 66,000 synthetic spectra for training, 396 spectra for validation, and 1,300 observed spectra for testing. The SpeX spectra has a resolving power of $R \approx 120$ and cover 0.89–2.45 μm [2,3]. Each spectrum was represented by a 90-feature vector combining normalized spectral bins, targeted spectral slopes, *J*- and *H*-band magnitudes and field-referenced residual features. Random Forest, Support Vector Machine, K-Nearest Neighbors, and a one-dimensional Convolutional Neural Network were evaluated [4]. Candidate configurations were selected using the validation set, with dwarf-type balanced accuracy [5], while final performance was assessed exclusively on the test set.

The following equations define the slope-based and field-referenced features.

$$R_i(\lambda_k) = \frac{F_i(\lambda_k)}{F_{\text{field},\,t_i}(\lambda_k)} - 1,$$

$$A_i = \text{median}_{\lambda_k \in \Lambda_i} |R_i(\lambda_k)|,$$

$$Y_i = \frac{1}{N_{\text{YJ},\,i}} \sum_{\lambda_k \in R_{\text{YJ}}} R_i(\lambda_k), \quad H_i = \frac{1}{N_{\text{H},\,i}} \sum_{\lambda_k \in R_{\text{H}}} R_i(\lambda_k), \quad K_i = \frac{1}{N_{\text{K},\,i}} \sum_{\lambda_k \in R_{\text{K}}} R_i(\lambda_k),$$

$$D_i = Y_i - \frac{H_i + K_i}{2},$$

$$S_{\lambda_1,\lambda_2} = \frac{F(\lambda_2) - F(\lambda_1)}{\lambda_2 - \lambda_1}.$$

R_i: residual relative to the same-subtype field standard; A_i: overall residual deviation; Y_i, H_i, K_i: regional mean residuals; D_i: YJ–HK contrast; $S_{\lambda 1,\lambda 2}$: spectral slope.

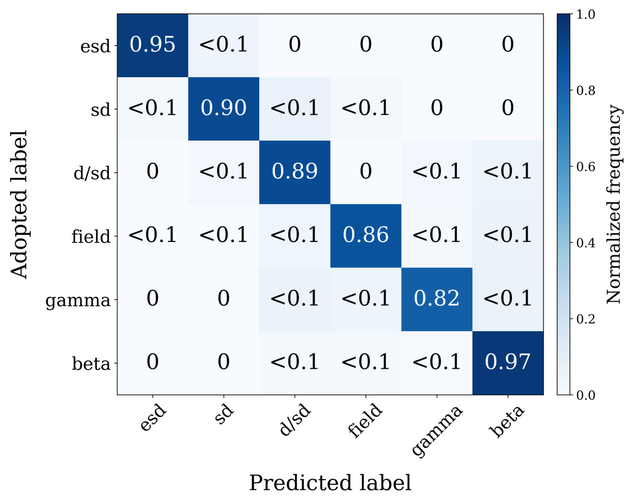


Fig. 2: Row-normalized KNN dwarf-type confusion matrix on the test set. Diagonal values represent the recall of each category.

RESULTS

KNN produced the most uniform recognition across the six dwarf-type categories, with class recalls ranging from 0.82 to 0.97. The highest recalls were obtained for *β* and esd objects, while *γ* objects remained the most difficult category (see Fig. 2).

Figure 3 identifies D, A, K, J, H, spectral slopes, and 0.89–0.95 μm bins as the most informative RF features.

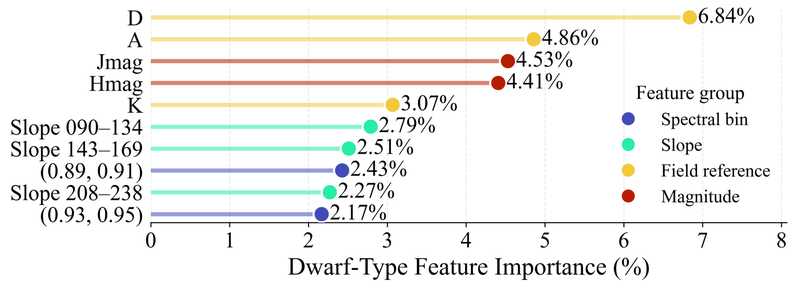


Fig. 3: Top 10 RF features ranked by normalized impurity-based importance for dwarf-type classification.

CONCLUSIONS

- KNN achieved the best class-balanced performance, with 89.84% dwarf-type BAcc and 96.85% spectral-type accuracy within ±1 subtype, above the 89.46% dwarf-type accuracy reported by Zhou et al. [1]
- Using dwarf-type BAcc reduced the influence of the dominant field class.
- *D, A, Jmag*, and *Hmag* added valuable information beyond the spectral bins.
- KNN produced the most uniform recall across the six classes, compared to RF, SVM and CNN.

REFERENCES

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